

Jing-Zhang Switchyard

THE SWITCHYARD

A City-AI Interchange on a Century-Old Railway

Memory · Living · Innovation Tracks — One Spine, Three Plazas, Five Wings

Centennial Jing-Zhang AI Innovation Belt Open Call · Formal Package (A3 Booklet)

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Provisional substitute boundary — display and design discussion only, not a redline

1 Master Concept: A City-Scale Switchyard

Reading the century-old Jing-Zhang Railway as a city-scale switchyard: talent, enterprises, data, scenarios and capital enter, switch tracks, go

Three tracks: Memory (century-old Jing-Zhang culture) / Living (urban AI lifestyle) / Innovation (AI convergence)

Five functions mapped to marshalling-yard roles: power depot (Zhongzhiyuan), marshalling yard (AI Origin Community), test track (Xiaoyu

Spatial structure: one spine (innovation spine) three plazas (Dazhongsi/Wudaokou/Zhongzhiyuan) five wings (R&D/living/green/culture/co

Three pilgrimage landmarks: Origin Signal Tower / Interchange Hall City-Agent Console / Switch Monument Open-Source Pulse Ring

Fig.1 Jing-Zhang Switchyard master concept (provisional boundary)

Jing-Zhang Switchyard THE SWITCHYARD • Master Concept & Spatial Structure

One spine · three plazas · five wings · Memory / Living / Innovation tracks · provisional boundary (dashed)



Source: provisional_boundaries.geojson & submission geometry (displayed EPSG:4326, recalculated EPSG:4548). Provisional boundary — display and design discussion only, not a redline.

2 Three-Level Scope & Land-Use Structure

Coordinated research 43.6 km² · overall design 11.4 km² (submission boundary) · key areas 368.4 ha (three areas)

Land-use mix (EPSG:4548): R&D/commercial/residential/culture/education/park green/buffer green/plaza/reserved/road

Green ratio 35.3% · Public space ratio 7.6% · Building density 14.4% · Concept FAR 0.72

Fig.2 Land-use zones & functional structure (seamless full partition)

Land-use Zones & Functional Structure

Full partition within the submission boundary · seamless & overlap-free · EPSG:4548 recalculated · non-statutory land use



Source: geometry/land_use.geojson (EPSG:4548). Provisional boundary for display and design discussion only.

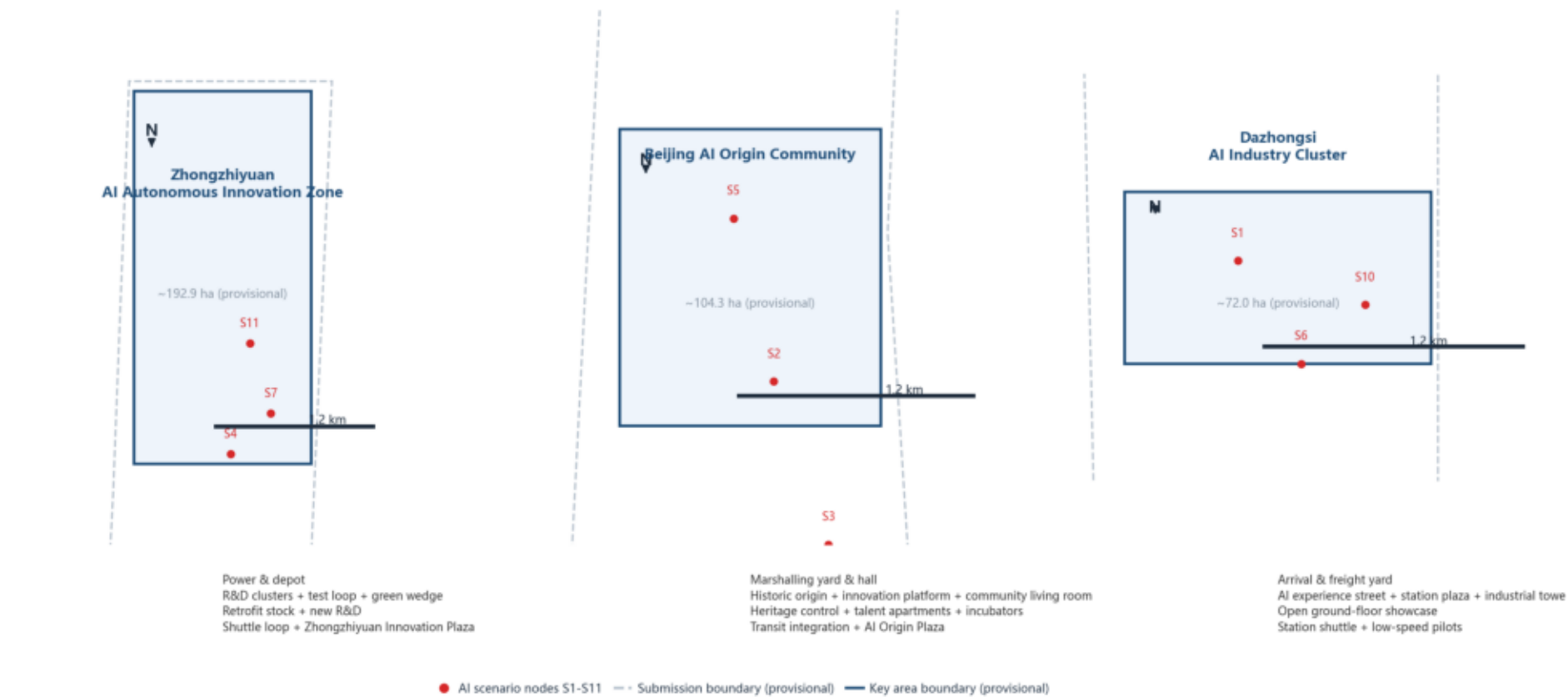
3 Key Area Detailed Design

Zhongzhiyuan·power depot (192.1ha): R&D clusters+test loop+green wedge; AI Origin Community·marshalling yard (104.3ha): historic origin+in

Fig.3 Key area index (provisional boundary)

Key Area Detailed Design Index

Provisional boundary · design discussion only · per area: positioning + spatial structure + building renewal + mobility + public space + AI scenarios + implementation risks



All key areas use provisional polygons for directional design only; implementation risks (5th Ring/Qinghe ecology, heritage scope, property rights, transit integration) require official confirmation.

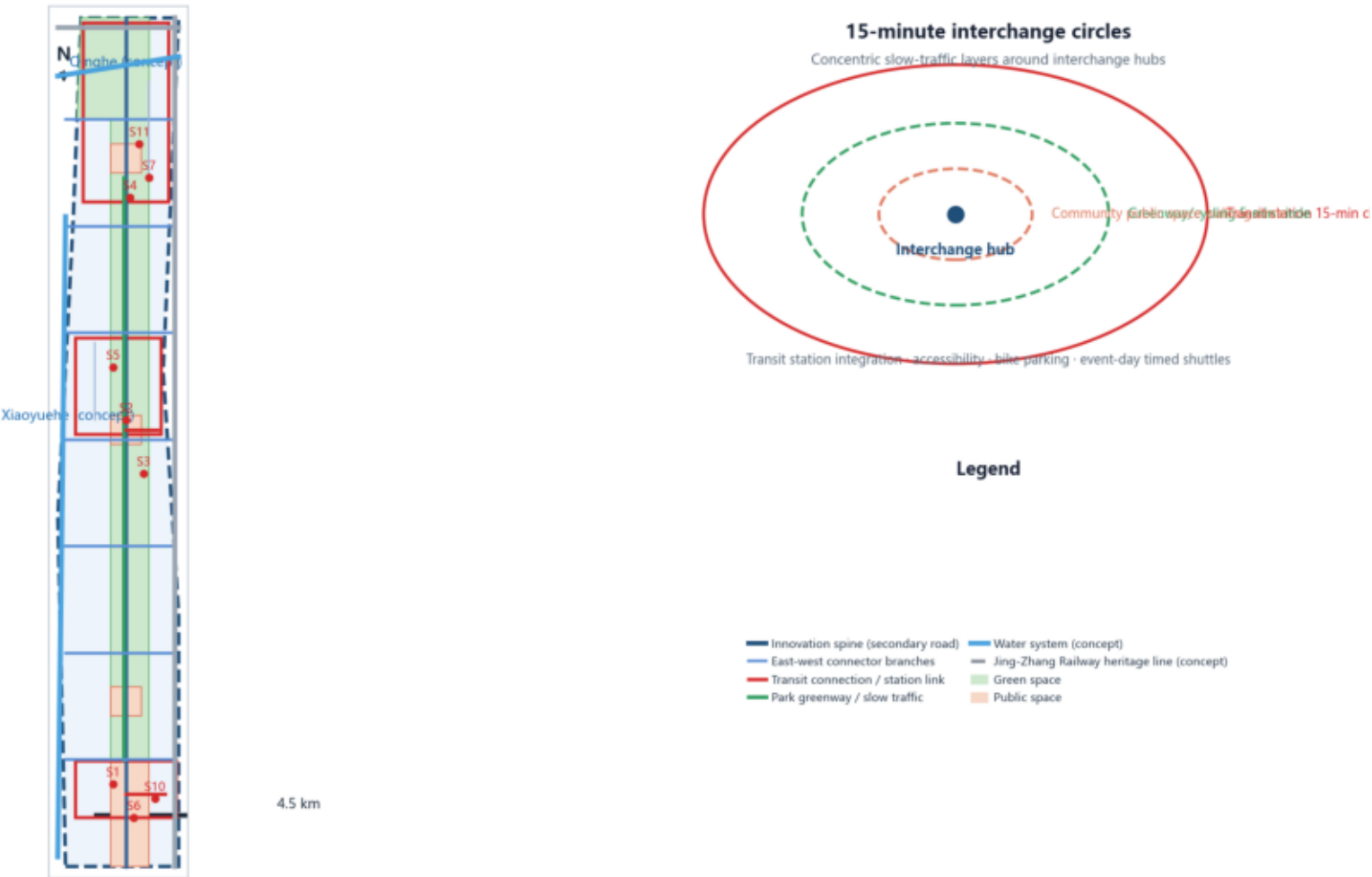
4 Mobility & Blue-Green Public Space

Innovation spine + transit links + park greenway; one vertical (Jing-Zhang heritage park vitality belt) two horizontal (Qinghe, Xiaoyuehe) b

Fig.4 Mobility & blue-green public space system (concept centerlines, not redlines)

Mobility & Blue-Green Public Space System

Interchange first · one spine two wings blue-green network · concept road centerlines (not redlines) · transit links & low-speed pilots



Roads are concept centerlines, not redlines; water and rail are provisional concept locations. Source: geometry/roads, green_space, public_space, constraints.

5 Regional Synergy & AI Innovation Ecosystem

Left: overall spatial structure + regional innovation synergy (Beiwei/Future Science City/Huairou Science City/E-Town/Beijing-Tianjin-Hebei); Right: AI in

Concept-level directions only; no enterprise lists, investment or policy commitments



Fig.5 Overall spatial structure + regional innovation synergy (concept)

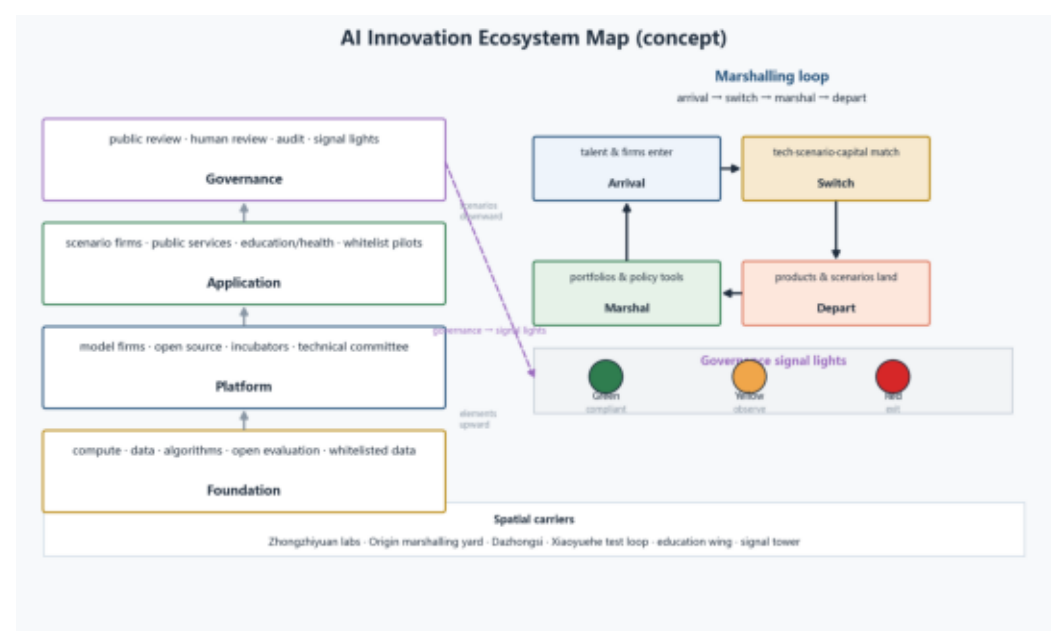


Fig.6 AI innovation ecosystem map (concept)

6 Core Metrics & Implementation Phasing

Site 11,412,825 m² · Scenario nodes 12 · Concept clusters 6 · Road ratio 4.9%

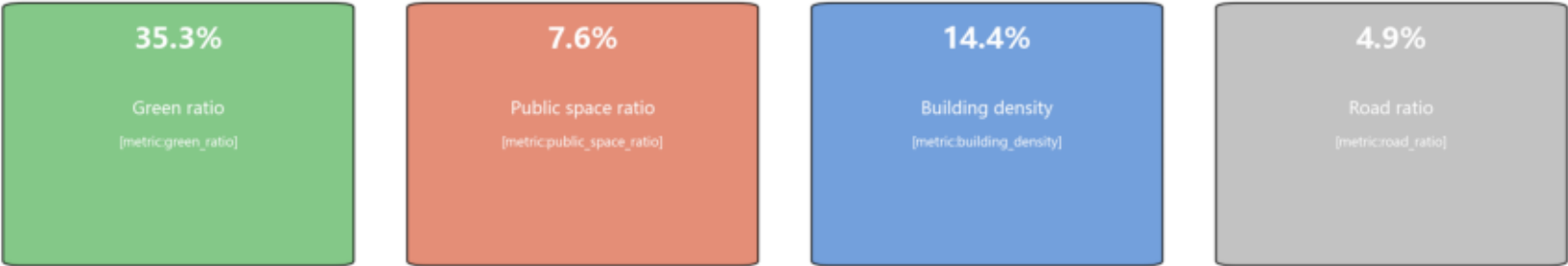
Near (1-3y) 4,696,434 m² · Mid (3-5y) 4,161,324 m² · Far (5-10y) 2,555,092 m²

Fig.7 Core metrics recalculation & evidence chain (EPSG:4548)

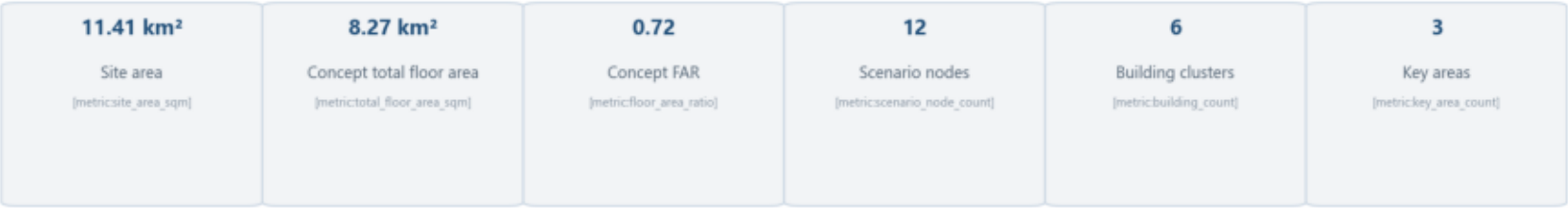
Core Metrics Recalculation & Evidence Chain

All spatial metrics recalculated from geometry in EPSG:4548 · concept values to be checked against regulatory controls

I. Key Ratio Indicators (main)



II. Core Data (auxiliary)



III. Phasing & Evidence Formulas

Near phase_1 4696 ha · Mid phase_2 4161 ha · Far phase_3 2555 ha
site_area=polygon_area(site_boundary) · green_ratio=green/site · public_ratio=public/site · FAR=total_floor/site · density=footprint/site · road_ratio=Σ(L×W)/site
Formulas and source_files see metrics.json; all areas are concept recalculations under the provisional boundary and must be recalculated once the official boundary is available.